
The Gut–Brain Axis and Depressive Disorders: A Prospective Cohort Study of Gut Microbiome Composition, Inflammatory Biomarkers, and Clinical Outcomes

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ABSTRACT

Background: Converging evidence from animal models and small human studies suggests that gut microbiome composition influences brain function and mood through the bidirectional gut–brain axis. However, large-scale prospective data relating specific microbial taxa to incident depression and treatment response remain scarce. **Methods:** We conducted a 24-month prospective cohort study enrolling 1,248 adults (18–65 years) across four clinical sites. Shotgun metagenomic sequencing of stool samples was performed at baseline, 6, 12, and 24 months. Depressive symptom burden was assessed with the PHQ-9 at each visit, with DSM-5 major depressive disorder (MDD) diagnosis confirmed by structured clinical interview. Serum inflammatory biomarkers (IL-6, TNF- α , CRP, LPS-binding protein) were measured at each timepoint. **Results:** Participants who developed incident MDD ($n = 187$; 15.0%) showed significantly lower baseline abundance of *Lactobacillus reuteri*, *Faecalibacterium prausnitzii*, and *Akkermansia muciniphila* (all $q < 0.05$ after FDR correction), alongside elevated IL-6 and CRP. A microbiome-derived dysbiosis index (MDI) we developed predicted 12-month MDD onset with AUC = 0.74 (95% CI: 0.69–0.79). Among antidepressant-treated participants, baseline *F. prausnitzii* abundance was the strongest independent predictor of SSRI response (OR = 2.31, 95% CI: 1.44–3.71). **Conclusions:** These findings support a prospective link between gut microbiome dysbiosis, systemic inflammation, and depressive disorders, and suggest that microbiome profiling may improve risk stratification and treatment personalisation in psychiatry.

Keywords: gut–brain axis, microbiome, major depressive disorder, metagenomics, inflammation, SSRI response, dysbiosis index

1. Introduction

Major depressive disorder (MDD) affects approximately 280 million people worldwide and is a leading cause of disability-adjusted life years (GBD Collaborators, 2022). Despite decades of research into monoaminergic and neuroplasticity-based mechanisms, roughly one third of patients fail to respond to

first-line antidepressant therapy (Cipriani et al., 2018). This treatment gap has intensified interest in alternative biological frameworks, among which the gut–brain axis has emerged as particularly promising.

The gut–brain axis comprises neural (vagus nerve), endocrine (HPA axis, enteroendocrine hormones), immune (cytokines, pattern recognition receptors), and metabolic (short-chain fatty acids, tryptophan metabolites) pathways through which gut microbiota communicate bidirectionally with the central nervous system (Cryan et al., 2019). Germ-free rodent experiments have established causal roles for gut bacteria in anxiety-like behaviour, HPA axis reactivity, and hippocampal neurogenesis (Sudo et al., 2004). Human cross-sectional studies have identified reduced *Lactobacillus* and *Bifidobacterium* abundance in depressed patients (Simpson et al., 2021), and a randomised controlled trial found that multi-strain probiotic supplementation modestly reduced PHQ-9 scores versus placebo (Schaub et al., 2022).

Nevertheless, the existing human literature is predominantly cross-sectional, limited by small sample sizes and inconsistent microbiome profiling methods. Critically, prospective data linking baseline microbiome composition to incident MDD and longitudinal treatment outcomes are largely absent. The present study addresses this gap through a large, multi-site prospective cohort design with repeated shotgun metagenomic sampling and comprehensive clinical phenotyping.

2. Methods

2.1 Study Design and Participants

The Gut–Brain Prospective (GBP) cohort enrolled adults aged 18–65 years at four sites: Kyoto University Hospital, Johns Hopkins Hospital, Hospital Universitario La Paz (Madrid), and the Wellcome Sanger clinical research unit (Cambridge, UK). Exclusion criteria included: current MDD at baseline, use of antibiotics or probiotics within 3 months of enrolment, inflammatory bowel disease, immunosuppressive therapy, and pregnancy. Written informed consent was obtained from all participants; the study was approved by the ethics committees of all four institutions (UMIN000054321).

2.2 Microbiome Profiling

Stool samples were self-collected using OMNIgene-GUT kits and shipped on ice to a central laboratory at the Wellcome Sanger Institute within 48 hours. DNA was extracted using the QIAamp PowerFecal Pro DNA Kit. Shotgun metagenomic libraries were prepared with the Nextera XT kit and sequenced on Illumina NovaSeq 6000 (2 × 150 bp paired-end, 20 Gb per sample). Taxonomic profiling was performed with MetaPhlAn 4.0 against the CHOCOPhIAn database; functional profiling used HUMAnN 3.0. Alpha diversity was quantified as Shannon entropy and observed species richness; beta diversity was assessed using Bray–Curtis dissimilarity with PERMANOVA.

2.3 Clinical Assessments

Depressive symptom burden was quantified with the Patient Health Questionnaire (PHQ-9) at baseline and 6, 12, and 24-month visits. Incident MDD was defined as PHQ-9 ≥ 10 at two consecutive visits plus DSM-5 criteria confirmed by the Mini International Neuropsychiatric Interview (MINI 7.0) administered

by trained clinicians. Among participants initiating antidepressant therapy, SSRI response was defined as $\geq 50\%$ reduction in PHQ-9 score at 12 weeks.

3. Results

3.1 Cohort Characteristics

Of 1,248 enrolled participants, 1,211 (97.0%) completed at least one follow-up visit and were included in primary analyses. Mean age was 38.4 ± 12.1 years; 54.6% were female. Over 24 months, 187 participants (15.0%) met criteria for incident MDD. Incident MDD cases did not differ significantly from non-cases in age, sex, BMI, or antibiotic use during follow-up (all $p > 0.10$).

Table 1. Baseline characteristics by incident MDD status.

Characteristic	No MDD (n=1024)	Incident MDD (n=187)	p-value
Age, mean $\pm$ SD (years)	38.2 ± 12.3	39.1 ± 11.4	0.38
Female sex, n (%)	560 (54.7%)	102 (54.5%)	0.97
BMI, mean $\pm$ SD (kg/m ²)	24.1 ± 3.8	24.6 ± 4.1	0.11
Shannon entropy, mean $\pm$ SD	3.61 ± 0.44	3.29 ± 0.51	<0.001
IL-6 (pg/mL), median [IQR]	2.1 [1.3–3.4]	3.8 [2.2–6.1]	<0.001
CRP (mg/L), median [IQR]	1.4 [0.7–2.9]	2.7 [1.3–5.4]	<0.001
PHQ-9 baseline score	2.3 ± 1.9	3.1 ± 2.2	<0.001

3.2 Microbiome Predictors of Incident MDD

Linear discriminant analysis effect size (LEfSe) identified 23 taxa with significantly different baseline abundance between incident MDD cases and non-cases. The three most depleted species in future MDD cases were *Faecalibacterium prausnitzii* (LDA score = 3.71), *Akkermansia muciniphila* (LDA score = 3.44), and *Lactobacillus reuteri* (LDA score = 3.02). Conversely, *Clostridium bolteae* and members of the genus *Ruminococcus* were enriched in future MDD cases.

Our microbiome-derived dysbiosis index (MDI), computed as a weighted log-ratio of the top 10 depleted versus top 10 enriched taxa, predicted 12-month MDD onset with AUC = 0.74 (95% CI: 0.69–0.79) in leave-one-site-out cross-validation. The MDI remained a significant predictor after adjusting for baseline PHQ-9 score, IL-6, CRP, age, sex, and site (adjusted OR per SD increase = 1.88, 95% CI: 1.41–2.51, $p < 0.001$).

3.3 Microbiome and Antidepressant Response

Among 214 participants initiating SSRI therapy during follow-up, 12-week response data were available for 196 (91.6%). SSRI responders ($n = 118$, 60.2%) had significantly higher baseline *F. prausnitzii* relative abundance than non-responders (median 2.41% vs 1.07%, Mann–Whitney U $p < 0.001$). In multivariable logistic regression, each doubling of baseline *F. prausnitzii* abundance was associated with 2.31-fold higher odds of SSRI response (OR = 2.31, 95% CI: 1.44–3.71, $p < 0.001$), independent

of baseline PHQ-9 severity, age, sex, and site.

4. Discussion

This prospective multi-site cohort study provides the largest-scale evidence to date that gut microbiome composition at baseline prospectively predicts incident MDD and antidepressant treatment response. The depletion of *F. prausnitzii*, *A. muciniphila*, and *L. reuteri* in future MDD cases replicates findings from smaller cross-sectional studies (Cryan et al., 2019; Simpson et al., 2021) and extends them to a prospective context, strengthening causal inference.

The mechanistic basis for these associations likely involves multiple pathways. *F. prausnitzii* is a major producer of butyrate, a short-chain fatty acid that maintains gut epithelial integrity, modulates microglial activation, and promotes hippocampal BDNF expression in rodent models. *A. muciniphila* preserves mucus layer integrity and reduces LPS translocation, dampening peripheral and neuroinflammation. *L. reuteri* produces indole-3-aldehyde, an aryl hydrocarbon receptor ligand with immunoregulatory properties, and influences vagal afferent signalling.

Clinically, the MDI's AUC of 0.74 for 12-month MDD prediction, while not yet at the threshold for standalone clinical utility, suggests genuine discriminative value that could augment standard risk assessment tools. The association between *F. prausnitzii* abundance and SSRI response is particularly intriguing given that SSRIs have well-documented immunomodulatory properties; it is plausible that a pro-inflammatory gut environment blunts the therapeutic effect of serotonin reuptake inhibition.

5. Conclusions

Gut microbiome dysbiosis, characterised by depletion of butyrate-producing and mucus-preserving taxa, prospectively predicts incident MDD and SSRI treatment response in a large multi-site cohort. These findings open the door to microbiome-based stratification in psychiatric care and suggest that adjunctive microbiome-targeted interventions (dietary fibre, targeted probiotics, fecal microbiota transplantation) warrant evaluation in randomised controlled trials of antidepressant augmentation.

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